
Capital Gateway

Full-Stack Website Development Report

Prepared by:	Arshin Salim
Agency:	CreativeInk
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Frontend repo:	https://github.com/arshinanu/capital-gateway
Backend repo:	https://github.com/arshinanu/pipedrive-timeline
Frontend host:	Vercel (React SPA)
Backend host:	Docker container — Spring Boot 3 / Java 21

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1. Executive Summary

This report provides a comprehensive account of the complete technical design, development, and deployment of the **Capital Gateway** digital platform — comprising both the customer-facing website and the backend integration service that powers it. Capital Gateway is a UK FCA-authorised independent finance broker, and the platform serves as both a marketing and operational tool: attracting new clients, facilitating loan applications, automating CRM data entry, and managing secure document submissions.

The platform is split into two independently deployed services: a **React SPA frontend** hosted on **Vercel** at **capitalgatewayltd.com**, and a **Spring Boot backend** API (crm-bridge) deployed as a Docker container. The backend acts as an integration hub, bridging the website with Pipedrive CRM, TimelinesAI WhatsApp messaging, Microsoft OneDrive document storage, and GoDaddy email.

4 Web Pages	35+ Finance Products	400+ Lender Network	5 Integrated Services
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Key Deliverables

- Marketing website presenting 35+ specialist finance products across 4 lending categories
- Interactive repayment calculator for property and business loan scenarios
- Three-step guided loan application form with Pipedrive CRM integration and bank statement upload
- Secure, adviser-initiated document portal for KYC and financial document submission
- Passwordless magic-link client authentication via email
- Spring Boot 3 backend service (Java 21, WebFlux) handling all API, CRM, and messaging logic
- Automated WhatsApp welcome messages via TimelinesAI on new application submission
- Scheduled stale-deal follow-up system (cron, Mon–Fri 9am) sending WhatsApp nudges
- Dual cloud file storage: bank statements saved to both Pipedrive and Microsoft OneDrive
- Pipedrive webhook handler triggering WhatsApp messages on deal stage changes
- Docker-based backend deployment with full environment-variable configuration
- Full mobile responsiveness with WCAG-considerate, accessible design

2. Project Overview

Capital Gateway is an independent, FCA-authorised finance brokerage connecting UK businesses and property owners to a panel of 400+ specialist lenders across business lending, bridging finance, development finance, and commercial and residential mortgages.

2.1 Business Context

Attribute	Detail
Business Name	Capital Gateway
Type	Independent Finance Broker (FCA Authorised)
Target Market	UK Businesses, SMEs, Property Investors & Developers
Production URL	https://www.capitalgatewayltd.com
Phone	07538 346 421
Email	Info@capitalgatewayltd.com
Office	2 Leyson Road, Cheltenham, GL51 6RU
Lender Panel	400+ specialist and mainstream UK lenders
Avg. Decision	48 hours
Finance arranged	£500M+

2.2 Project Objectives

- **Marketing & Lead Generation** — Present the full service range compellingly to prospective clients
- **Client Conversion** — Convert visitors into qualified enquiries via a frictionless multi-step form
- **Operational Efficiency** — Automatically pipe new enquiries into Pipedrive CRM, eliminating manual data entry
- **Instant Client Communication** — Send personalised WhatsApp welcome messages the moment an application is submitted
- **Document Management** — Provide a branded, secure portal for KYC/financial document collection
- **Automated Follow-Up** — Proactively re-engage stale deals without manual intervention
- **Brand Differentiation** — Establish a premium, trustworthy brand presence in the UK brokerage market
- **Regulatory Transparency** — Clearly communicate FCA authorisation and maintain compliance in all visible content

3. Technology Stack

3.1 Frontend Stack

Category	Technology	Version	Purpose
Runtime	React	18.3.x	Component-based UI framework
Language	JavaScript (JSX)	ES2022+	Primary development language
Build Tool	Vite	5.3.x	Fast bundling, HMR, production builds
Routing	React Router DOM	7.14.x	Client-side SPA navigation
Animation	Framer Motion	12.38.x	Declarative UI animations and transitions
Icons	Lucide React	0.400.x	Consistent SVG icon library
Hosting	Vercel	—	Global CDN, HTTPS, SPA routing

3.2 Backend Stack

Category	Technology	Version	Purpose
Language	Java	21 (LTS)	Primary backend language
Framework	Spring Boot	3.3.4	Application framework and auto-configuration
HTTP Layer	Spring WebFlux	6.x	Reactive, non-blocking HTTP server (Netty)
HTTP Client	WebClient	6.x	Non-blocking HTTP client for external APIs
Validation	Spring Validation	3.x	Bean validation for request DTOs
Email	Spring Mail	3.x	SMTP email dispatch (magic-link auth)
Build	Maven	3.9.x	Dependency management and build lifecycle
Utilities	Lombok	—	Boilerplate reduction
Testing	JUnit 5 + Reactor Test	—	Unit and reactive stream testing
Containerisation	Docker	—	Container image build and runtime
Orchestration	Docker Compose	—	Local and server orchestration

4. Frontend Application Architecture

4.1 Routing

Route	Component	Purpose
/	HomePage	Full marketing landing page
/apply	ApplyPage	Multi-step loan enquiry and application form
/login	LoginPage	Client portal magic-link authentication
/upload	DocumentUploadPage	Secure KYC document submission portal

4.2 Component Hierarchy

Level	Components
App (Root)	Routes → HomePage · ApplyPage · LoginPage · DocumentUploadPage
Page Layout	Nav · Footer · PrivacyPolicy (modal overlay)
Home Sections	Hero · LendersStrip · Services · About · Calculator · Testimonials · TrustpilotBanner · LatestNews · Contact
Page Forms	Registration (3-step) · DocumentUpload form · Login form
Sub-components	VideoScrollBackground · StepIndicator · Field · ReviewBlock · ReviewRow

4.3 API Contract — Frontend Calls to Backend

Endpoint	Method	Caller	Payload
/api/loan-applications	POST	Registration	multipart/form-data: JSON data + bank statement files
/api/enquiries	POST	Contact	application/json: name, email, phone, loanType, amount, message
/api/auth/login	POST	LoginPage	application/json: email
/api/upload	POST	DocumentUpload	multipart/form-data: documentType + file(s) + email query param

The backend base URL is injected at build time via the `VITE_API_URL` environment variable, allowing the same frontend codebase to target different backends across development, staging, and production environments.

5. Website Pages & User Journeys

5.1 Home Page (/)

Section	Anchor	Purpose
Hero	#top	First impression: headline, video background, key stats, primary CTA
Lenders Strip	—	Social proof: scrolling animation of 400+ lender names
Services	#services	Product range: accordion with 4 categories and 35+ products
About	#about	Trust building: company values and differentiators
Calculator	#calculator	Engagement: interactive loan repayment estimator
Testimonials	#testimonials	Social proof: client success stories
Trustpilot Banner	—	Third-party review rating display
Latest News	#news	Content marketing: industry news
Contact	#contact	Conversion: direct enquiry form and contact details

5.2 Apply Page (/apply) — Three-Step Application

Step	Label	Fields Collected
1	Contact	First/last name, email, phone (intl. dialling), company name, business structure, industry / purchase vehicle, time trading
2	Finance	Finance product, loan amount, urgency, purpose, turnover / property details, bank statements (file upload), notes
3	Review	Full summary for user verification — submits as multipart/form-data POST to backend on confirm

5.3 Login Page (/login)

Implements passwordless magic-link authentication. The user submits their email; the backend verifies the email against Pipedrive, then dispatches a login email via the GoDaddy SMTP service. The UI always shows a confirmation screen on submission — preventing email enumeration attacks.

5.4 Document Upload Page (/upload)

Feature	Detail
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Document Types	Proof of Identity, Proof of Address, Bank Statements, Payslips, P60/SA302, Employment Contract, Mortgage Statement, Other
Accepted Formats	PDF, JPG, JPEG, PNG, WEBP
Max File Size	10 MB per file
Upload Method	Drag-and-drop zone or file browser (multiple files)
Client Identification	Email pre-filled from adviser-issued URL query parameter (?email=...)
Validation	MIME type and size validated client-side before upload
Terms Consent	Mandatory Privacy & Cookie Policy checkbox before submission
Backend Action	Finds most-recently-updated Pipedrive deal for client email; uploads files to that deal

6. Service Offerings

Capital Gateway's product range spans four lending categories with 35+ individual products, presented in an interactive accordion on the website — each product accessible via a detail modal overlay with a direct link to the contact section.

Business & SME Loans (11 products)

Product
Merchant Cash Advance
Working Capital
Business Term Loan
Asset Finance
Invoice Finance
Factoring
Funding for Stock
Overdraft / Revolving Credit
Business Acquisition Finance
Retail Furniture Finance
Loans Against Assets

Bridging Finance — Commercial & Residential (10 products)

Product
Quick Bridge (24/48 hr)
Standard Auction Bridge Loan
Residential Bridge Loan
Commercial Bridge Loan
Regulated Bridge Loan
Unregulated Bridge Loan
Refurbishment Loan
Bridge to Let
Development Exit Finance

Re-Bridge / Safe Exit

Development Finance (5 products)

Product
Purchase Land With or Without Planning
Development Exit Finance
First Time Developer Finance
Experienced Developer Finance
Refurbishment / Value-Addition Finance

Mortgage — Commercial & Residential (6 products)

Product
Purchase of Commercial Property
Purchase of Semi-Commercial Property
Loans for Purchasing Assets & Fittings
Loans for Refurbishment
Commercial Re-Mortgage
Commercial Development Finance

7. Key Interactive Features

7.1 Navigation

- Fixed navigation with scroll-aware frosted-glass transition (transparent → dark blur after 30px scroll)
- Smooth-scroll anchor links to homepage sections
- Accent-coloured 'Apply' CTA and pill-styled 'Login' link
- Hamburger menu collapsing to full-screen mobile overlay on viewports below 900px
- Framer Motion logo entrance animation with hover scale

7.2 Hero Section

- Full-viewport height with autoplay muted video background
- Left-to-right directional gradient overlay preserving video on wider screens
- Staggered word-by-word headline animation on page load
- Four key stats (400+ lenders, 48hr decision, 100% transparent fees, FCA) with perspective hover lift
- Floating animated cards: '48h avg decision' card and '£500M+ arranged' badge
- Ambient rotating 3D wireframe shapes with prefers-reduced-motion support
- Pulsing green status pill 'Accepting new clients'

7.3 Services Accordion

- Click-to-expand accordion using CSS grid row animation (0fr → 1fr) — no JS height measurement
- Animated green left-border active indicator; icon and tag update colour on open state
- Sub-items in an auto-fill responsive grid with 3D perspective hover lift
- Product detail modal with spring zoom-in animation, backdrop blur, and Escape/click-outside close
- 'Discuss this option' CTA scrolls to contact section

7.4 Interactive Loan Calculator

Input	Control	Range
Loan Type	Toggle button	Property / Business
Loan Amount	Range slider	£50k – £10M
Term	Range slider	1 – 40 years
Income / Turnover	Range slider	£30k – £2M
Indicative Rate	Editable inline display	Default: 5.25% (property) / 6.85% (business)

Results are computed client-side with React useMemo (zero-latency). Output includes: monthly payment, total interest, debt-to-income ratio, total repayable, and an eligibility indicator (DTI threshold 40%). Custom rates persist to localStorage between sessions.

8. Backend Architecture — CRM Bridge Service

The backend is a standalone **Spring Boot 3** microservice written in **Java 21**, built with **Maven** and named *crm-bridge* (version 0.1.0). It is deployed as a **Docker container** on port 8080 and acts as the integration layer between the Capital Gateway website, Pipedrive CRM, TimelinesAI WhatsApp, Microsoft OneDrive, and GoDaddy email.

The service uses **Spring WebFlux** with Project Reactor for a fully non-blocking, reactive programming model — all I/O operations (HTTP calls to Pipedrive, TimelinesAI, Microsoft Graph, and SMTP) are handled as reactive pipelines using **Mono** and **Flux**, maximising throughput with a small thread pool.

8.1 Package Structure

Package	Contents
com.example.crmbridge	CrmBridgeApplication — Spring Boot entry point
com.example.crmbridge.config	Configuration property classes: PipedriveProperties, TimelinesAiProperties, OneDriveProperties, FollowUpProperties, HttpClientConfig, WebConfig (CORS)
com.example.crmbridge.webhook	REST controllers: LoanApplicationController, EnquiryController, LoginController, DocumentUploadController, PipedriveWebhookController, TimelinesAiWebhookController, ManualSendController, LeadController, DealLookupController, HealthController
com.example.crmbridge.pipedrive	PipedriveService (API client), PipedriveDtos (request/response records)
com.example.crmbridge.timelinesai	TimelinesAiService (WhatsApp API client), TimelinesAiDtos
com.example.crmbridge.onedrive	OneDriveService (Microsoft Graph file upload via OAuth2 client credentials)
com.example.crmbridge.email	EmailService (SMTP magic-link email dispatch)
com.example.crmbridge.scheduler	FollowUpScheduler (scheduled stale-deal WhatsApp nudges)

8.2 Reactive Programming Model

All controllers return `Mono<ResponseEntity>` or `Flux` types, processed by the Netty event loop in Spring WebFlux. The application makes heavy use of reactive chaining:

- `.flatMap()` — sequential dependent API calls (create org → create person → create lead → attach note → upload files → send WhatsApp)
- `.thenReturn()` — pass through the deal object after side-effect completions

- `.onErrorResume()` — graceful degradation for non-critical steps (file uploads, WhatsApp) so a single failure does not abort the whole pipeline
- `Mono.when()` — parallel fan-out for concurrent uploads to Pipedrive and OneDrive simultaneously
- `retryWhen(Retry.backoff())` — transient error retry with exponential backoff for TimelinesAI API calls
- `Schedulers.boundedElastic()` — offloads blocking SMTP calls to a dedicated thread pool

8.3 Configuration

All sensitive credentials and tunable parameters are externalised as environment variables. The application.yml uses Spring's `${ENV_VAR:defaultValue}` syntax for all external secrets:

Property	Env Variable	Purpose
<code>pipedrive.domain</code>	<code>PIPEDRIVE_DOMAIN</code>	Pipedrive company subdomain
<code>pipedrive.api-token</code>	<code>PIPEDRIVE_API_TOKEN</code>	Pipedrive API authentication token
<code>pipedrive.trigger-stage-id</code>	<code>PIPEDRIVE_TRIGGER_STAGE_ID</code>	Deal stage ID that triggers WhatsApp follow-up (default: 5)
<code>timelinesai.token</code>	<code>TIMELINESAI_TOKEN</code>	TimelinesAI API Bearer token
<code>timelinesai.sender-phone</code>	<code>TIMELINESAI_SENDER_PHONE</code>	Connected WhatsApp number for outbound messages
<code>followup.stale-after-days</code>	<code>FOLLOWUP_STALE_DAYS</code>	Days without activity before follow-up (default: 7)
<code>followup.cron</code>	<code>FOLLOWUP_CRON</code>	Cron expression for scheduler (default: Mon-Fri 9am)
<code>onedrive.tenant-id</code>	<code>ONEDRIVE_TENANT_ID</code>	Azure AD tenant for Microsoft Graph auth
<code>onedrive.client-id</code>	<code>ONEDRIVE_CLIENT_ID</code>	Azure AD app client ID
<code>onedrive.client-secret</code>	<code>ONEDRIVE_CLIENT_SECRET</code>	Azure AD app client secret
<code>onedrive.drive-id</code>	<code>ONEDRIVE_DRIVE_ID</code>	OneDrive drive ID from Microsoft Graph
<code>onedrive.folder-path</code>	<code>ONEDRIVE_FOLDER_PATH</code>	Target folder path (default: 'Bank Statements')
<code>spring.mail.password</code>	SMTP password in <code>.env</code>	GoDaddy SMTP credential for <code>info@capitalgatewayltd.com</code>

8.4 Email Configuration

Outbound emails are sent via GoDaddy's SMTP service using SSL on port 465. The from address is `info@capitalgatewayltd.com`. The email service runs on a bounded elastic thread pool to avoid

blocking the Netty event loop.

Setting	Value
Host	smtpout.secureserver.net
Port	465
Auth	SMTP AUTH (username/password)
Security	SSL (SSLSocketFactory)
From Address	info@capitalgatewayltd.com

9. Backend API Endpoints

All public-facing endpoints are under the `/api` base path. Webhook receivers are under `/webhooks`. The server runs on port 8080. Multipart file uploads are supported up to 20 MB per file and 100 MB per request.

9.1 POST `/api/loan-applications`

Controller: `LoanApplicationController` | **Content-Type:** `multipart/form-data` | **Called by:** Registration form (frontend)

Receives the full loan application including optional bank statement file attachments. Executes a sequential reactive pipeline:

- Parse the JSON data part into a `LoanApplicationRequest` record
- Create or skip an Organisation in Pipedrive (if company name is present)
- Create a Person in Pipedrive with name, email, and phone number
- Create a Lead in Pipedrive linked to the person, with the loan value as the deal value
- Attach a structured note to the deal with all additional finance fields (product, urgency, turnover, property details, exit strategy, LTV, etc.)
- Upload bank statement files in parallel to both Pipedrive (as deal file attachments) and Microsoft OneDrive (under 'Bank Statements' folder, prefixed with the loan title)
- Send a personalised WhatsApp welcome message via TimelinesAI to the applicant's mobile number
- Return 201 Created with the Pipedrive Deal object

9.2 POST `/api/enquiries`

Controller: `EnquiryController` | **Content-Type:** `application/json` | **Called by:** Contact form (frontend)

Creates a Person in Pipedrive, creates a Lead titled '[Name] - [Type] Enquiry', and attaches a note with loan type, amount, and any message text. Currency symbols and commas are stripped from the amount field before parsing to a numeric value.

9.3 POST `/api/login`

Controller: `LoginController` | **Content-Type:** `application/json` | **Called by:** Login page (frontend)

Searches Pipedrive for a Person with the submitted email address. If found, dispatches a login notification email via the `EmailService`. Returns 404 if no matching person exists. The frontend always shows a confirmation screen regardless of the result to prevent email enumeration.

9.4 POST `/api/upload`

Controller: DocumentUploadController | **Content-Type:** multipart/form-data | **Called by:** Document upload portal (frontend)

Accepts the client email as a query parameter and one or more files as multipart parts. Queries Pipedrive for all deals associated with that email address, selects the most recently updated deal, and uploads each file to that deal's file attachments in Pipedrive. Returns a success response listing the uploaded filenames and deal information.

9.5 POST /webhooks/pipedrive

Controller: PipedriveWebhookController | **Registered in:** Pipedrive webhook settings

Receives deal lifecycle events from Pipedrive and handles two triggers:

- **Deal created (action: create)** — Fetches the linked person's phone number and sends a WhatsApp welcome message referencing the deal title and website URL
- **Deal stage changed (action: change)** — If the deal has just entered the configured trigger stage (default: stage ID 5), sends a personalised WhatsApp follow-up message. Ignores all other stage transitions
- Responds immediately with 200 OK and processes the event asynchronously (`.subscribe()`) to prevent Pipedrive webhook timeout

9.6 POST /webhooks/timelinesai

Controller: TimelinesAiWebhookController | **Registered in:** TimelinesAI webhook settings (message:new / chat:incoming:new events)

Receives incoming WhatsApp reply events from TimelinesAI (when a client responds to an outbound message). Logs the event type, phone number, and message text. Provides an extension point for future automation such as updating Pipedrive notes or marking activities complete.

9.7 GET /health

Controller: HealthController | Returns 200 OK with a simple status payload. Used by infrastructure monitoring and Docker health checks.

10. Third-Party Integrations

The backend acts as a central integration hub connecting four external services. All API calls are made reactively via Spring WebClient with appropriate error handling and retry logic.

10.1 Pipedrive CRM

Role: Primary CRM and deal management platform. All lead data collected via the website is structured and persisted in Pipedrive.

Operation	Pipedrive API Call	Used By
Create Organisation	POST /organizations	LoanApplicationController
Create Person	POST /persons	LoanApplicationController, EnquiryController
Find Person by Email	GET /persons/search	LoginController, DocumentUploadController
Get Person by ID	GET /persons/:id	PipedriveWebhookController, FollowUpScheduler
Create Lead	POST /leads	LoanApplicationController, EnquiryController
Attach Note to Lead	POST /notes	LoanApplicationController, EnquiryController
Upload File to Deal	POST /files (multipart)	LoanApplicationController, DocumentUploadController
Find Deals by Email	GET /deals (filter by person)	DocumentUploadController
Find Stale Open Deals	GET /deals (status=open, sorted)	FollowUpScheduler

10.2 TimelinesAI — WhatsApp Business Messaging

Role: WhatsApp Business messaging platform. TimelinesAI is connected to Pipedrive and automatically logs all outbound WhatsApp messages as activities on the matching person's deal timeline — providing a complete communication audit trail without any additional Pipedrive API calls.

Attribute	Detail
API Endpoint	https://app.timelines.ai/integrations/api/messages
Authentication	Bearer token (TIMELINESAI_TOKEN)
Sender Number	Configured UK WhatsApp number (TIMELINESAI_SENDER_PHONE: +447587592759)

Message Triggers	New loan application, new deal (Pipedrive webhook), deal stage change, stale deal follow-up
Retry Policy	2 retries with exponential backoff (1s base); 4xx errors are not retried
CRM Logging	Automatic — TimelinesAI syncs all messages as Pipedrive deal activities

10.3 Microsoft OneDrive (Microsoft Graph API)

Role: Secondary cloud document storage. Bank statement files uploaded via the loan application form are saved to a dedicated 'Bank Statements' folder in OneDrive simultaneously with being attached to the Pipedrive deal. This provides a redundant, searchable document archive independent of Pipedrive.

Attribute	Detail
API	Microsoft Graph API v1.0 — /drives/{driveId}/root/{path}/content
Authentication	OAuth2 Client Credentials (Azure AD app registration)
Token Endpoint	https://login.microsoftonline.com/{tenantId}/oauth2/v2.0/token
File Naming	'{Loan Title} - {original filename}' (special characters sanitised)
Target Folder	'Bank Statements' (configurable via ONEDRIVE_FOLDER_PATH)
Fail Behaviour	onErrorResume — upload failure is logged but does not abort the loan application

10.4 GoDaddy SMTP — Email Delivery

Role: Transactional email delivery for the magic-link authentication flow. Spring Mail sends via GoDaddy's SMTP relay. The email notifies the client that a login request was received for their account.

Attribute	Detail
SMTP Host	smtpout.secureserver.net
Port / Security	465 / SSL
From	info@capitalgatewayltd.com
Use Case	Login notification email sent when a client requests portal access
Threading	Offloaded to Schedulers.boundedElastic() — non-blocking

11. Automated Workflows & Scheduling

The backend includes both event-driven and time-driven automation to ensure prospective clients receive timely, personalised communication throughout the application lifecycle — without any manual intervention from the broker.

11.1 Loan Application — Submission Workflow

Triggered when a prospect submits the Apply form on the website:

Step	Action	Service
1	Parse multipart form data	LoanApplicationController
2	Create Organisation in Pipedrive	PipedriveService
3	Create Person in Pipedrive	PipedriveService
4	Create Lead linked to Person	PipedriveService
5	Attach structured note to Lead	PipedriveService
6a	Upload bank statements to Pipedrive (parallel)	PipedriveService
6b	Upload bank statements to OneDrive (parallel)	OneDriveService
7	Send WhatsApp welcome message	TimelinesAiService
8	Return 201 Created + Deal JSON	LoanApplicationController

11.2 Pipedrive Webhook — Deal Stage Change Workflow

Triggered when Pipedrive fires a webhook on deal update:

Step	Action	Service
1	Receive webhook POST from Pipedrive	PipedriveWebhookController
2	Respond 200 OK immediately (async processing)	HTTP layer
3	Check: is entity type 'deal'?	PipedriveWebhookController
4a	If action = 'create': send welcome WhatsApp	TimelinesAiService
4b	If action = 'change' AND entered trigger stage: send follow-up WhatsApp	TimelinesAiService

11.3 Scheduled Follow-Up — Stale Deal Workflow

A Spring `@Scheduled` cron job runs every weekday at 9:00am (configurable via `FOLLOWUP_CRON`). It scans all open Pipedrive deals that have had no activity in the last 7 days (configurable via `FOLLOWUP_STALE_DAYS`) and sends a gentle WhatsApp nudge to each associated contact. Up to 4 messages are sent concurrently.

Step	Action	Service
1	Cron fires (Mon-Fri 9:00am)	FollowUpScheduler
2	Fetch all open Pipedrive deals with no activity > 7 days	PipedriveService
3	For each stale deal: fetch linked Person's phone number	PipedriveService
4	Send personalised 'circling back' WhatsApp message	TimelinesAiService
5	Log result; continue to next deal (concurrency: 4)	FollowUpScheduler

11.4 WhatsApp Message Templates

Trigger	Message Template
New application (LoanApplicationController)	Hi {firstName}, thank you for your {product} enquiry with Capital Gateway Ltd! We've received your application and a member of our team will be in touch shortly. For more information, visit www.capitalgatewayltd.com
New deal (Pipedrive webhook — create)	Hi {firstName}, welcome to Capital Gateway Ltd! We've received your enquiry about "{dealTitle}" and will be in touch with you shortly. For more information, visit www.capitalgatewayltd.com
Stage change (Pipedrive webhook — change)	Hi {firstName}, just following up on "{dealTitle}" — happy to answer any questions!
Stale deal (FollowUpScheduler)	Hi {firstName}, circling back on "{dealTitle}" — still open on your side?

12. Design & User Experience

12.1 Colour Palette

Variable	Hex	Usage
--ink	#1a2520	Body copy, headings, primary UI text
--accent (green)	#00e87a	CTAs, active states, stat highlights, indicators
--paper (dark bg)	#090c0b	Hero, calculator, dark section backgrounds
--ivory (light bg)	#f5f2ec	Card and form backgrounds, light sections
--muted	#6b7c75	Subheadings, labels, de-emphasised text
--line	#d4cfc7	Borders, dividers, rule lines

12.2 Typography

Variable	Typeface	Usage
--font-display	Editorial display serif	Headline, section titles, large metric numbers
--font-body	Geometric sans-serif	Body copy, paragraphs, form fields, navigation
--font-mono	Monospace	Labels, tags, metadata, technical identifiers

12.3 Responsive Breakpoints

Breakpoint	Viewport	Key Adaptations
Mobile	< 480px	Single-column hero stats; reduced padding; full-width form fields
Tablet	< 900px	Hamburger navigation; hero floating cards hidden; single-column sections
Desktop	>= 900px	Two-column layouts; floating hero cards visible; full desktop nav
Wide	>= 1280px	Max content width 1280px enforced; generous padding

12.4 Accessibility

- Semantic HTML heading hierarchy (h1 → h2 → h3)
- ARIA labels on all interactive controls and modals
- aria-modal and role='dialog' on overlay components
- Keyboard navigation: Escape closes modals; Enter activates file drop zones

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- Decorative elements marked `aria-hidden='true'`
 - `prefers-reduced-motion` disables all ambient animations
 - Focus-visible outlines on custom checkbox controls

13. Security, Compliance & Data Handling

13.1 Data Flow Overview

Data	Origin	Destination	Transport
Application form fields	Frontend	Pipedrive CRM	HTTPS → backend → Pipedrive API
Bank statement files	Frontend	Pipedrive + OneDrive	HTTPS → backend → dual upload
Contact form enquiry	Frontend	Pipedrive CRM	HTTPS → backend → Pipedrive API
Client email (login)	Frontend	Email inbox + Pipedrive	HTTPS → backend → SMTP + Pipedrive read
Uploaded KYC documents	Frontend	Pipedrive deal files	HTTPS → backend → Pipedrive API
Interest rate (calculator)	Browser	Browser only (localStorage)	Never transmitted

13.2 Security Measures

- **HTTPS everywhere** — All traffic served over TLS via Vercel CDN (frontend) and reverse proxy (backend)
- **No client-side secrets** — API keys and backend URLs are server-side only; the frontend only knows the backend base URL
- **Environment variable isolation** — All credentials stored in environment variables; never hardcoded in source
- **Email enumeration protection** — Login page always shows confirmation regardless of whether the email exists in Pipedrive
- **Frontend file validation** — MIME type and file size checked before any upload begins
- **CORS configuration** — Backend WebConfig restricts cross-origin access to trusted frontend origins
- **Bean validation** — All API request DTOs use Jakarta Validation annotations (@NotBlank, @Email) enforced by Spring
- **Reactive error isolation** — onErrorResume prevents single integration failures from exposing stack traces or crashing the pipeline
- **Non-blocking SMTP** — Email dispatch offloaded to a thread pool; email failures are logged, not surfaced to the client

13.3 FCA & Regulatory Compliance

- FCA Authorised status prominently displayed in the hero section

-
- Calculator labelled 'Illustrative only' with full 'not an offer' disclaimer and mortgage repossession risk notice
 - Privacy Policy accessible from the footer on all pages and linked from the document upload consent checkbox
 - Application form includes explicit data usage consent notice before submission
 - No credit file searches or credit reference checks are performed by the website

14. Deployment & Infrastructure

14.1 Frontend — Vercel

The React SPA is deployed to **Vercel** and served via Vercel's global CDN. The production domain is <https://www.capitalgatewayltd.com>. All HTTP traffic is automatically redirected to HTTPS.

Config	Value	Purpose
buildCommand	npx vite build	Runs Vite production build, outputs to /dist
outputDirectory	dist	Location of compiled static assets
rewrite rule	/(.*) → /index.html	SPA fallback: all paths served to React Router
HTTPS	Automatic	TLS certificates managed by Vercel
CDN	Vercel Edge Network	Global asset delivery from nearest POP

14.2 Backend — Docker Container

The crm-bridge backend is packaged as a Docker image using a two-stage Dockerfile and deployed as a container on port 8080. Docker Compose is provided for both local development and server orchestration.

Stage	Base Image	Action
Build stage	maven:3.9-eclipse-temurin-21	Resolves dependencies offline, then runs mvn package -DskipTests to produce crm-bridge-0.1.0.jar
Runtime stage	eclipse-temurin:21-jre-alpine	Copies the JAR, creates a non-root 'app' user, exposes port 8080, runs the JAR

14.3 Environment Variables (Docker Compose)

The docker-compose.yml passes all runtime credentials as environment variables, reading from a host .env file. The container restarts automatically unless explicitly stopped (restart: unless-stopped).

14.4 System Architecture Diagram

The following table summarises the complete system component relationships:

Component	Technology	Role	Network
capitalgatewayltd.com	React + Vercel CDN	Customer-facing web application	Public HTTPS

CRM Bridge Backend	Spring Boot + Docker	API & integration hub	Private port 8080
Pipedrive CRM	SaaS (cloud)	Lead, person & deal management	Outbound HTTPS
TimelinesAI	SaaS (cloud)	WhatsApp Business messaging	Outbound HTTPS
Microsoft OneDrive	Microsoft 365 (cloud)	Document cloud archive	Outbound HTTPS (Graph API)
GoDaddy SMTP	SaaS (cloud)	Transactional email delivery	Outbound SMTP/SSL 465

15. Source Control & Repositories

Both projects are version-controlled with **Git** and hosted on **GitHub** under the arshinanu account. The default branch for both repositories is `master`.

Repository	URL	Contents
Frontend	https://github.com/arshinanu/capital-gateway	React SPA: all UI components, pages, styles, Vite config, Vercel deployment config
Backend	https://github.com/arshinanu/pipedrive-timeline	Spring Boot crm-bridge: controllers, services, config, Dockerfile, docker-compose, Maven POM

15.1 Frontend Repository — Recent Commits

Commit Hash	Message
e87d151	adding accepted terms and condition
d8de2f0	changing vercel.json
9c8ecb0	adding document upload controller
6c55a66	adding the changes
ab71a8f	adding changes for about section

15.2 Branch Strategy

- Both repositories use a single `master` branch for active development
- Deployment is triggered automatically by Vercel on push to `master` (frontend)
- Backend deployments are performed via Docker image rebuild and container restart
- Environment-specific configuration is managed via `.env` files (excluded from version control via `.gitignore`)

16. Conclusion & Recommendations

The Capital Gateway platform represents a complete, production-grade digital system for a specialist finance brokerage. The frontend delivers a visually distinctive and functionally rich client experience at <https://www.capitalgatewayltd.com>, while the backend crm-bridge service automates the full client intake workflow — from initial enquiry through to CRM entry, document storage, and personalised WhatsApp outreach.

16.1 Summary of Achievements

- Full-stack production platform live at capitalgatewayltd.com (frontend) + Docker backend
- 35+ finance products across 4 categories with interactive accordion and modal detail views
- Real-time loan calculator supporting property and business scenarios
- Three-step application form with Pipedrive CRM integration, bank statement upload to Pipedrive and OneDrive
- Passwordless magic-link authentication for returning client portal access
- Automated WhatsApp welcome messages sent immediately on every new application submission
- Pipedrive webhook handler sending WhatsApp messages on deal creation and stage transitions
- Scheduled Mon-Fri 9am stale-deal follow-up system — zero manual effort required
- Secure KYC document portal linked to specific Pipedrive deal records
- Non-blocking reactive backend handling concurrent API calls with graceful failure isolation
- Docker containerised backend with environment-variable-driven configuration
- Fully responsive frontend tested from mobile through wide desktop viewports

16.2 Recommendations for Future Enhancement

SEO & Server-Side Rendering

Migrate to Next.js for SSR/SSG to improve search engine indexability. The current React SPA is not crawlable by default search engines.

Backend Authentication

Add API key or JWT authentication to the backend endpoints to prevent unauthorised use by third parties.

Database Persistence

Introduce a lightweight database (e.g. PostgreSQL) for logging application submissions, follow-up attempts, and WhatsApp message history for audit purposes.

Analytics

Integrate privacy-compliant analytics (e.g. Vercel Analytics) to track conversion rates, calculator engagement, and form completion.

CI/CD Pipeline

Configure GitHub Actions to automate backend Docker image builds and deployments on push to master, matching the Vercel auto-deploy for the frontend.

Live Mortgage Rate API

Replace the static indicative calculator rates with a live rate feed API to improve accuracy and user trust.

Login Link Security

Implement time-limited, single-use token links for the magic-link login flow rather than simple email notifications.

WhatsApp Reply Automation

Extend the TimelinesAI webhook controller to automatically update Pipedrive notes or activities when a client replies via WhatsApp.

This report was prepared as a comprehensive technical and functional reference document covering both the Capital Gateway frontend website (capitalgatewayltd.com) and the crm-bridge backend service. All descriptions are derived from direct analysis of the project source code.

*Prepared by **Arshin Salim** at **CreativeInk** | Report generated: 29 June 2026*

Frontend: github.com/arshinanu/capital-gateway | Backend: github.com/arshinanu/pipedrive-timeline